

From provable security to practical failure

What cryptography enables - and what it cannot prevent - in distributed graph learning

Aubin Birba - aubin.birba@gmail.com

11/02/2026 - Cybersecurity research presentation @ Hitachi Energy Canada

whoami

Aubin Birba



- Security researcher, finishing a PhD at Université du Québec À Montreal (UQÀM)
- Masters in Cyberdefense and Information Security from Université Polytechnique Hauts-de-France
- Interested in secure computation, privacy-preserving and adversarial machine learning
- <https://birbaubin.github.io/>

Outline

- Crypto'Graph
 - Context : the modern data sharing landscape
 - Secure distributed graph analytics
 - State-of-the art
 - Our approach
 - Results

Outline

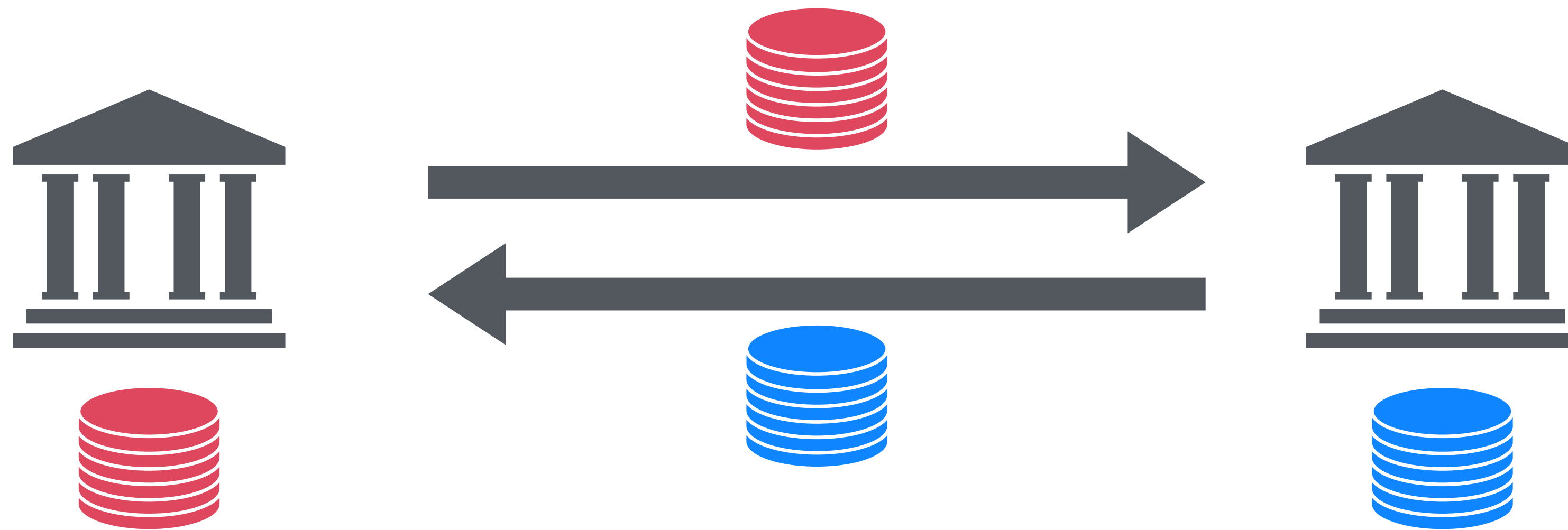
- GRAND
 - The limitations of cryptography
 - Reconstruction in the topological domain
 - Reconstruction in the spectral domain
 - Complete pipeline
 - Results
 - Cross-cutting insights
- Conclusion : relevance to power systems

Crypto'Graph : Leveraging Privacy-Preserving Link Prediction for Robust Graph Learning

The 14th ACM Conference on Data and Application Security and Privacy, 2024

Context

The modern data landscape



Data sharing is desirable for research and industrial advancement...

Context

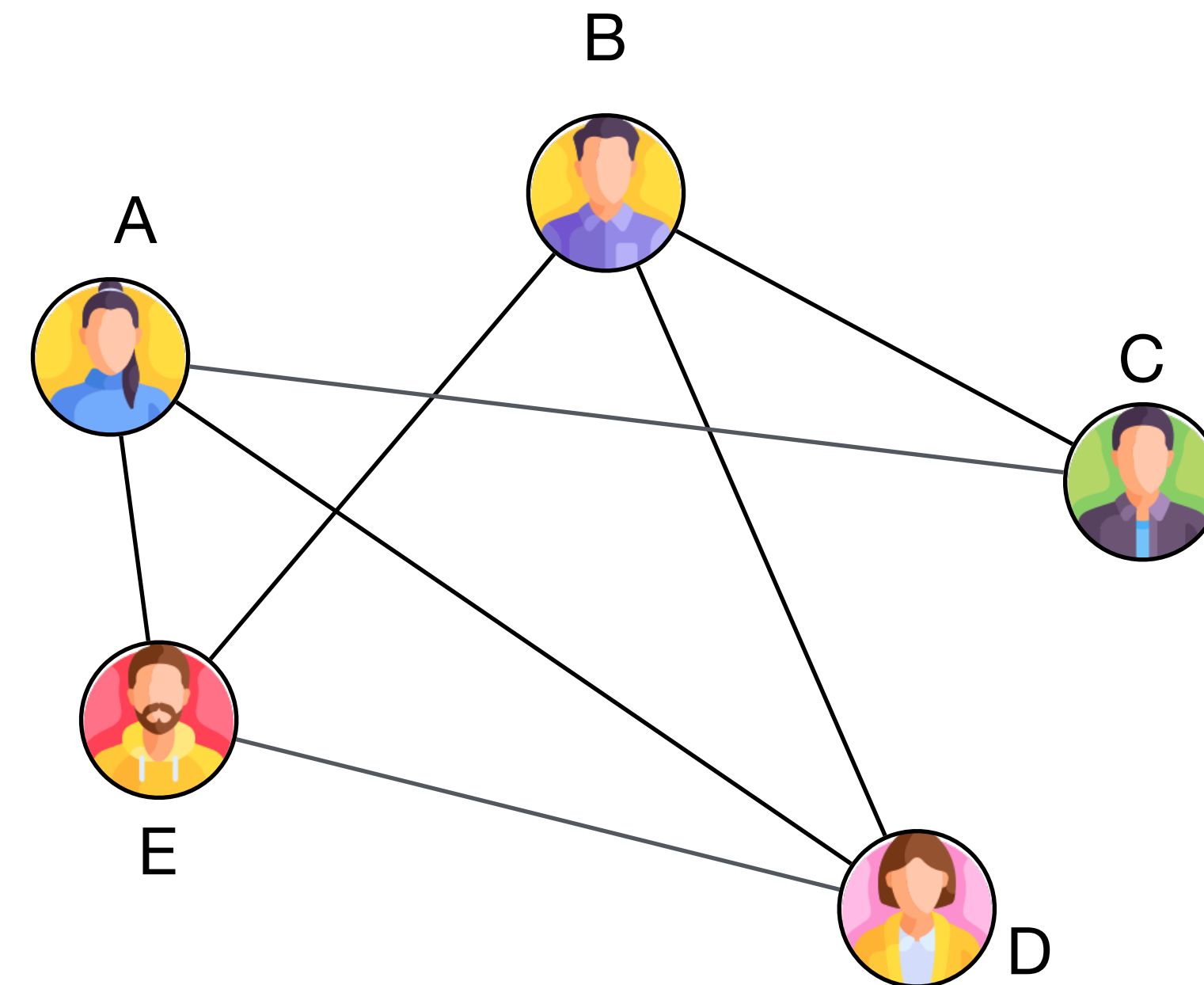
The modern data landscape



But it comes at a cost for confidentiality and
privacy
intellectual property
privacy regulation

Graph analytics

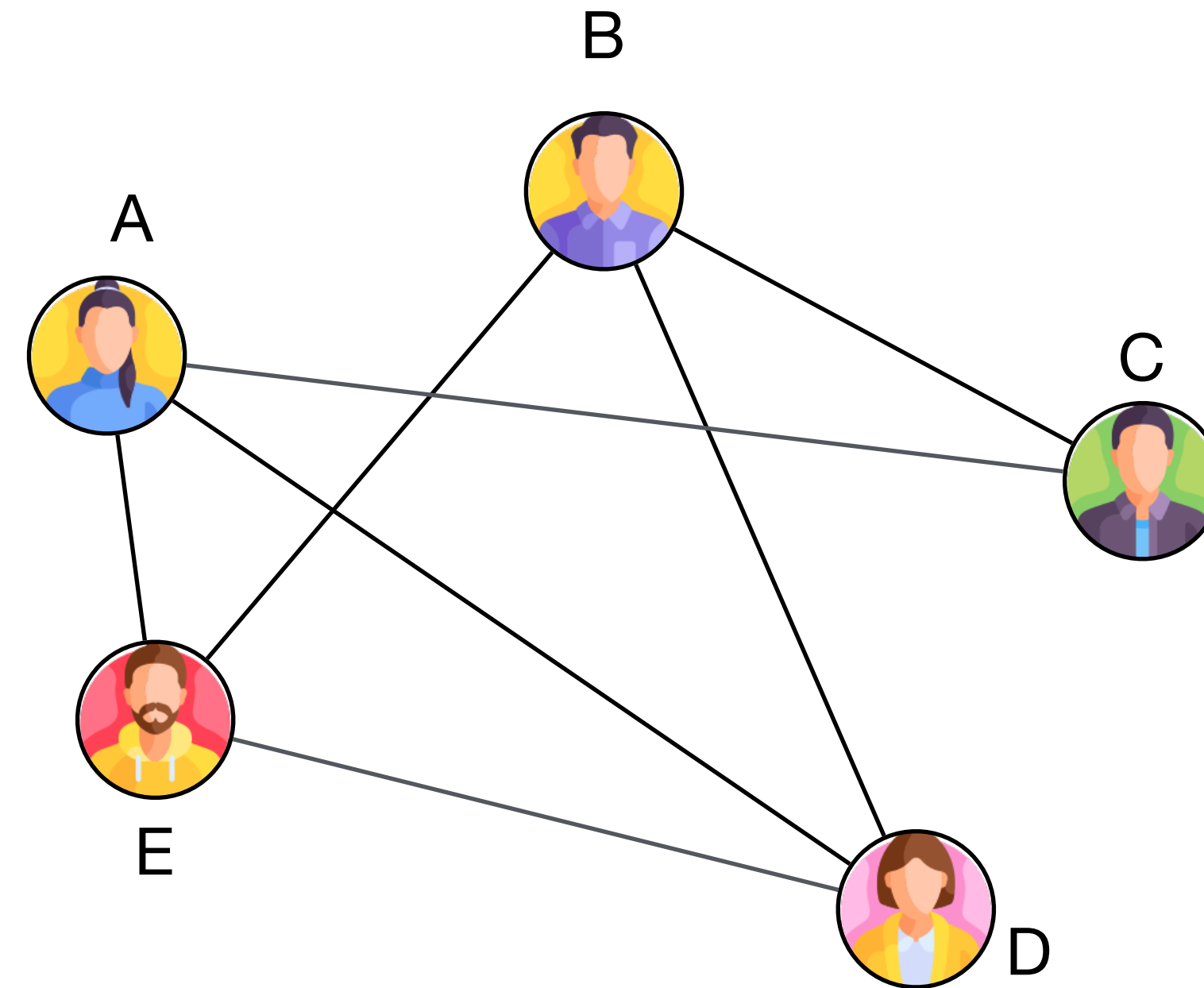
- Graphs are an important data structure that models relationships between entities (sensor networks, power networks)
- Graph analytics and Graph Machine Learning allow deep insights and exploitation of graphs for various purposes (topology growth forecasting, signal processing, etc)



Graph analytics

Link prediction

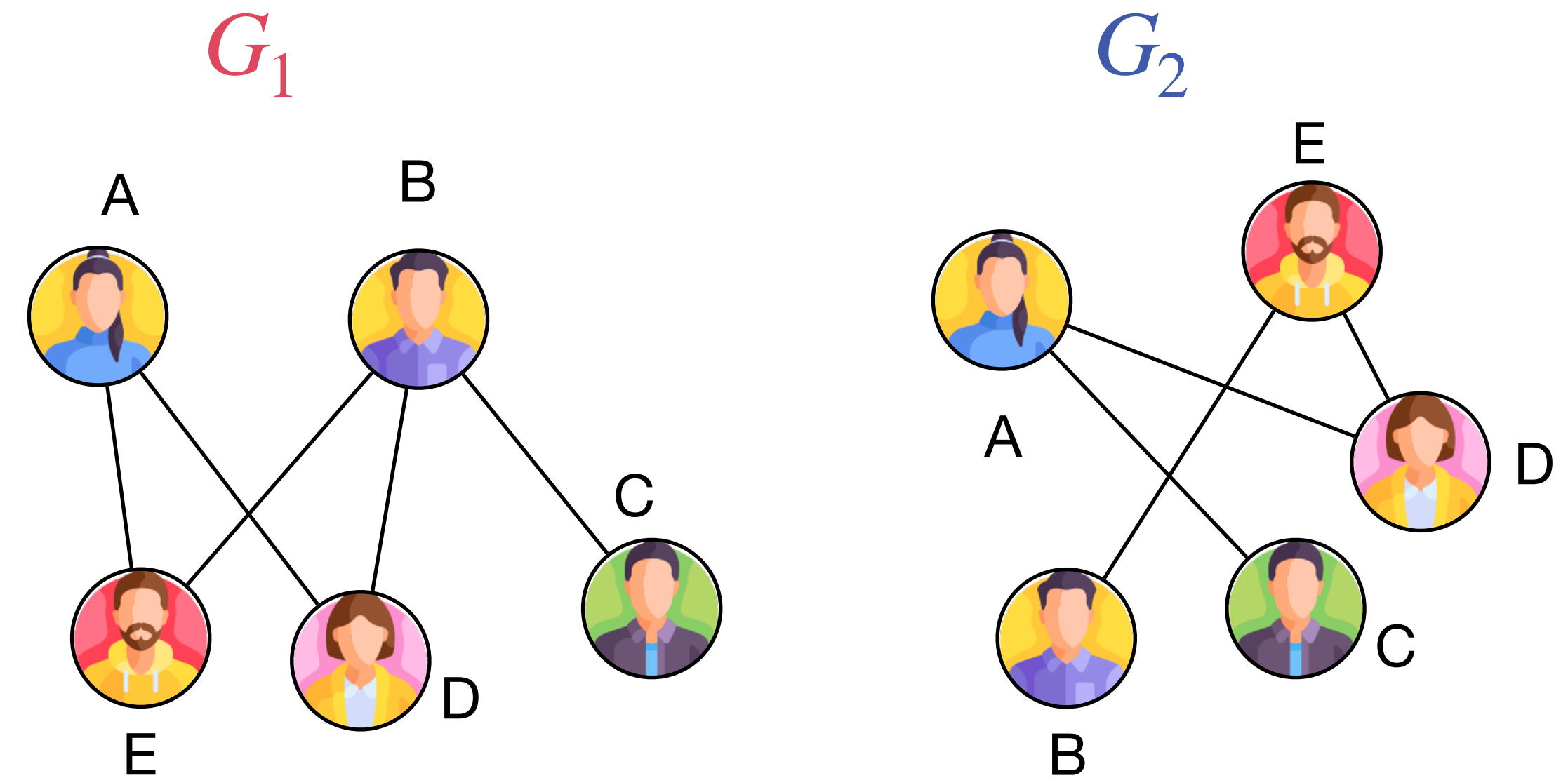
- Important graph analytics task that aims at identifying potential future connexions in the network (connection forecasting)
- A widespread approach to link prediction : similarity measures (similar nodes tend to form connections in many types of networks)
- Potential applications : network growth forecasting (*Zhang et al*), data curation (*Wu et al.*)



Secure distributed graph analytics

Requirements and constraints

- **System model** : Parties who hold graphs collaborate to perform link prediction on their joint network
- **Threat model** : each party tries to break the confidentiality of the other parties network edges while respecting the protocol (honest-but-curious parties)



Can we efficiently and privately,
predict links on a distributed graph ?

State of the art

	Accuracy	Speed	Cryptographic primitive	Similarity measures
Zhang et al.	Approximate	~1000s	Secret sharing	Common neighbours, Jaccard, Cosine, Adamic Adar
Demiral et al.	Exact	~15000s	PSI	Common neighbours
Crypto'Graph	Exact	~ 200s	PSI	Common neighbours, Jaccard, Cosine

Didem Demirag, Mina Namazi, Erman Ayday, and Jeremy Clark. Privacy preserving link prediction.. Data Privacy Management, Cryptocurrencies and Blockchain Technology. 2023

Hai-Feng Zhang, Xiao-Jing Ma, Jing Wang, Xingyi Zhang, Donghui Pan, and Kai Zhong. Privacy-preserving link prediction in multiple private networks. IEEE Transactions on Computational Social Systems. 2023

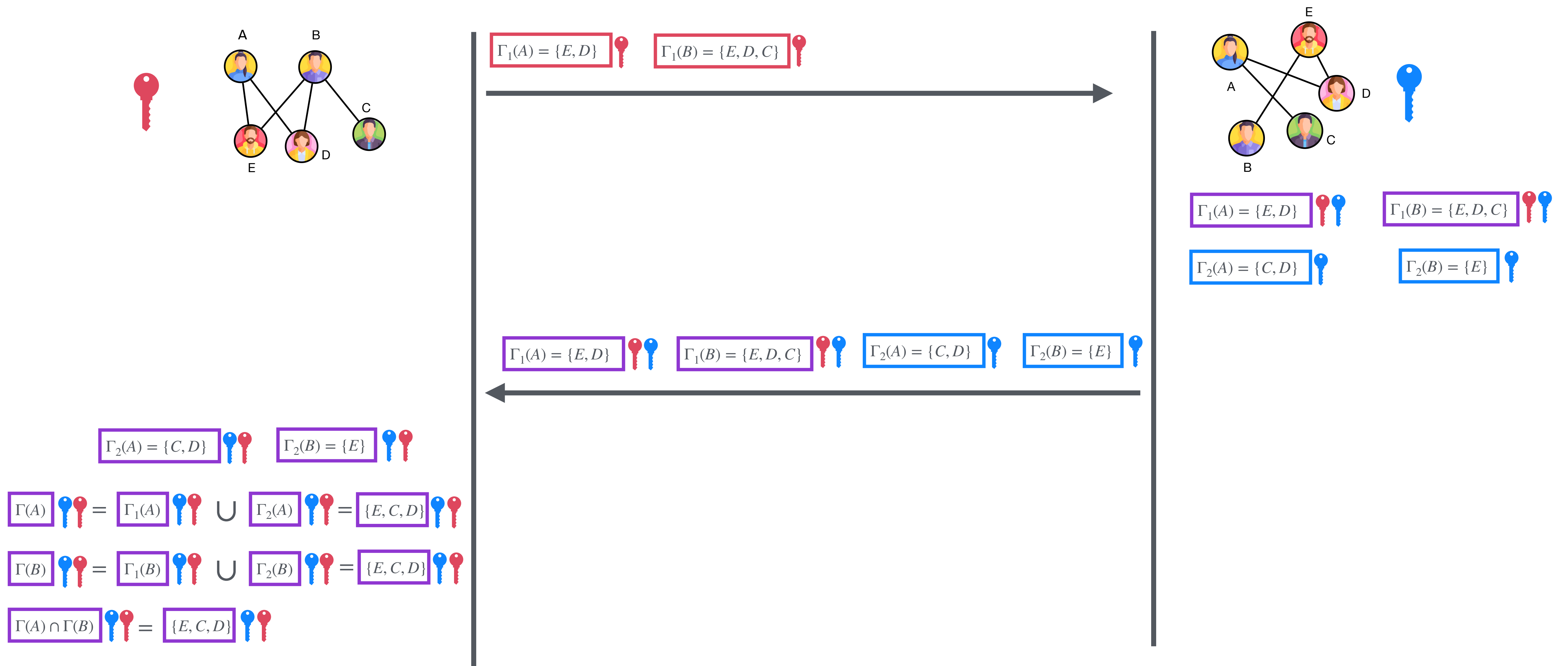
Azogagh, Sofiane, Zelma Aubin Birba, Sébastien Gambs, and Marc-Olivier Killijian. "Crypto'Graph: Leveraging Privacy-Preserving Distributed Link Prediction for Robust Graph Learning." In Proceedings of the Fourteenth ACM Conference on Data and Application Security and Privacy,. 2024.

Preliminaries

- G the distributed graph, V the set of vertices, E the set of edges
- G_1, G_2 the two partitions of the graph such that V belongs to both, and $E_1 \cup E_2 = E$
- $\Gamma(x)$ the neighbours of vertex x
- $\Gamma_i(x)$ the neighbours of vertex x on graph i

Crypto'Graph

Secure distributed link prediction



Crypto'Graph

Secure distributed link prediction

Let $\mathbb{G}$ be a cyclic group over $\mathbb{Z}_p$ (p a big prime number), and g a generator of this group

- Key generation : $\alpha, \beta \leftarrow \text{KeyGen}(1^\lambda)$
- Encryption : $\text{Enc}_k(x) = g^{kx}, k \in \{\alpha, \beta\}$
- Double encryption : $\text{Enc}_{k_1, k_2}(x) = (g^{k_1 x})^{k_2} = g^{k_1 k_2 x}$
- Discrete logarithm assumption : for p big enough, given (g, g^e) , it is hard to get e

Crypto'Graph

Results

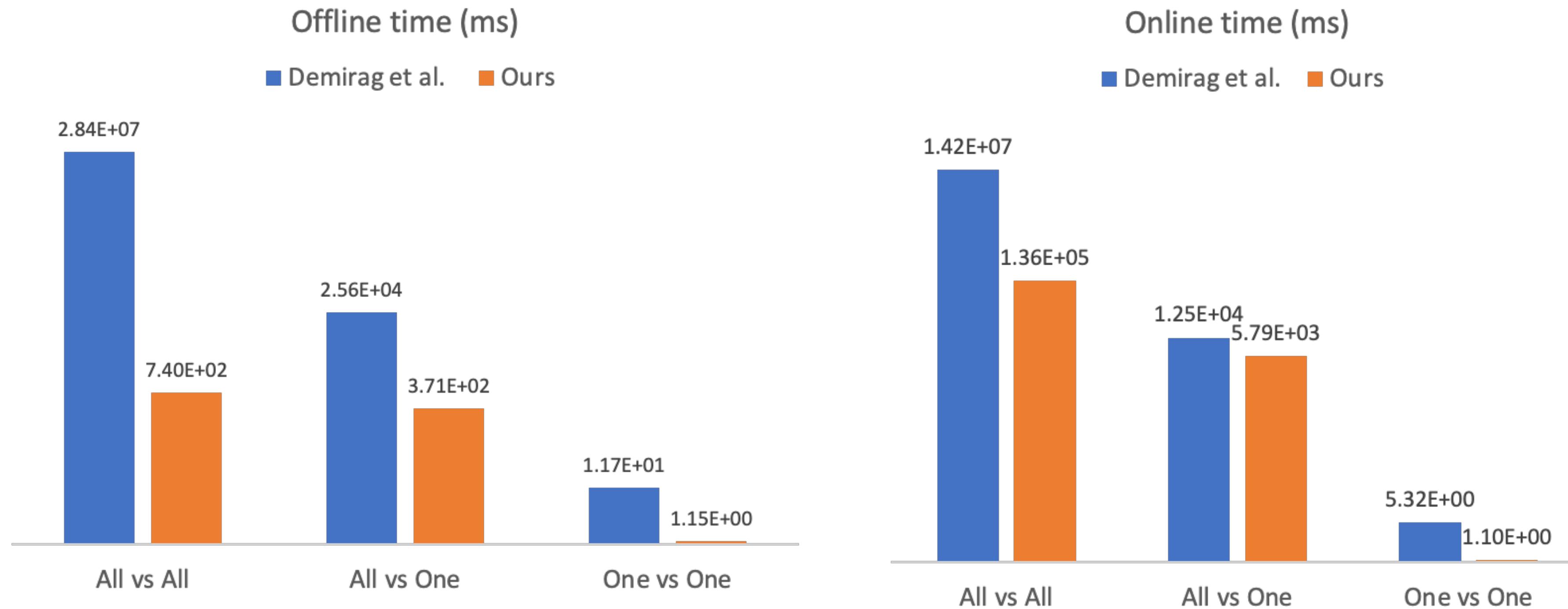
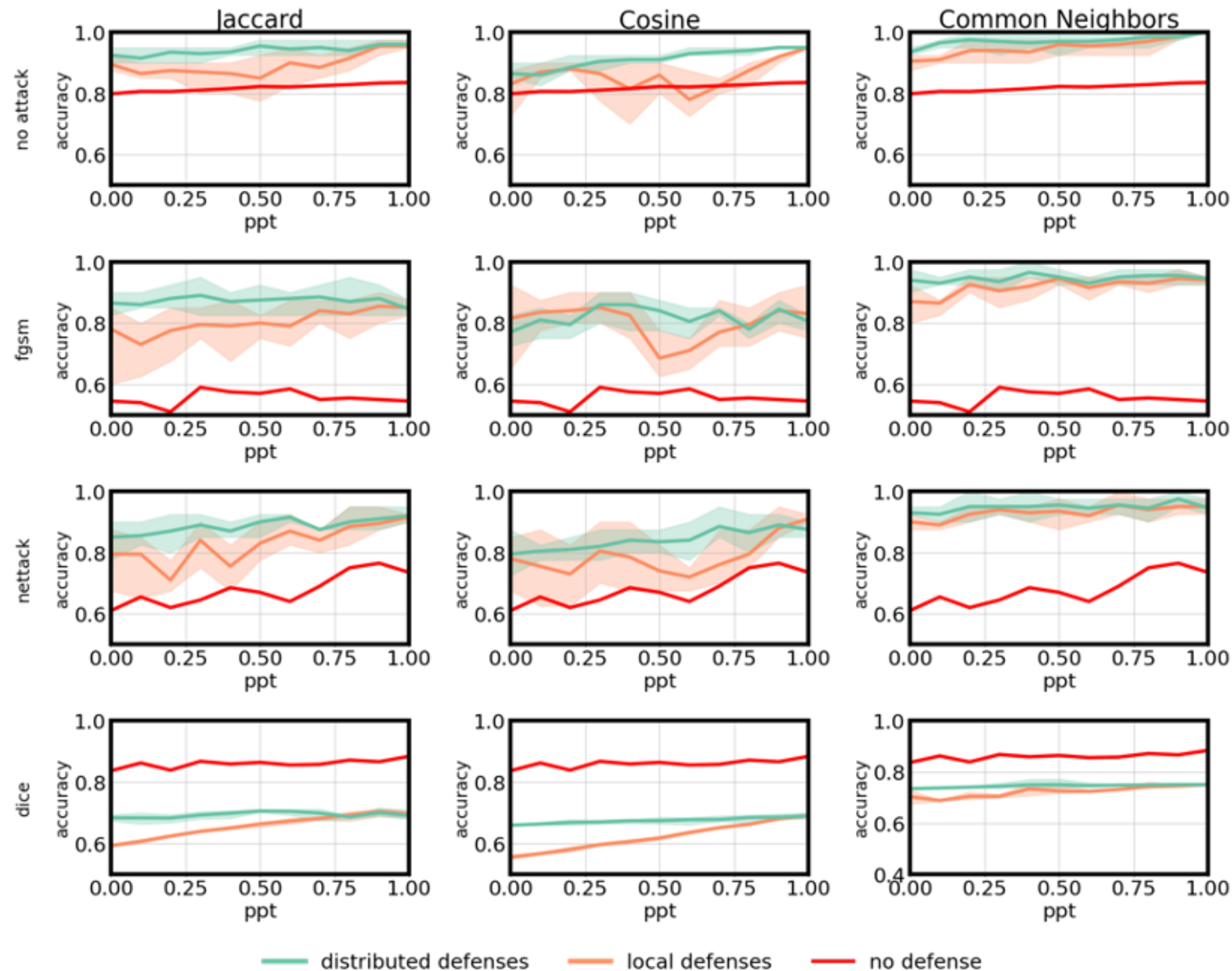


Figure 1 : Runtime off Crypto'Graph, compared to Demirag et al

- Link prediction for all node pairs (all vs all), between one node and all others (all vs one), and between two nodes (one vs one)
- We improve time complexity by orders of magnitude

Crypto'Graph

Results



- Crypto'Graph is used as a private preprocessing mechanism
- Links between nodes that have low similarity are considered erroneous and dropped
- Graph neural networks are trained on the resulting graphs
- Crypto'Graph based defense allows better accuracy than local and no defense

Crypto'Graph

What cryptography enables here

- We can efficiently compute the number of common neighbours on a distributed graph
- The common neighbours metric serves as a partial result for the computation of other similarity metrics (Jaccard Index, cosine similarity, preferential attachment index, etc)
- The structure of the graph is preserved by the Diffie-Hellmann like encryption
- **Proof shows that the protocol is secure as long as the Discrete Logarithm Assumption holds**

But what if ...

- We get the number of common neighbours for all the node pairs in the graph
- Can one of the parties infer some edges on the private graph of the other ?
-

GRAND : Graph Reconstruction from potential partial Adjacency and Neighborhood Data

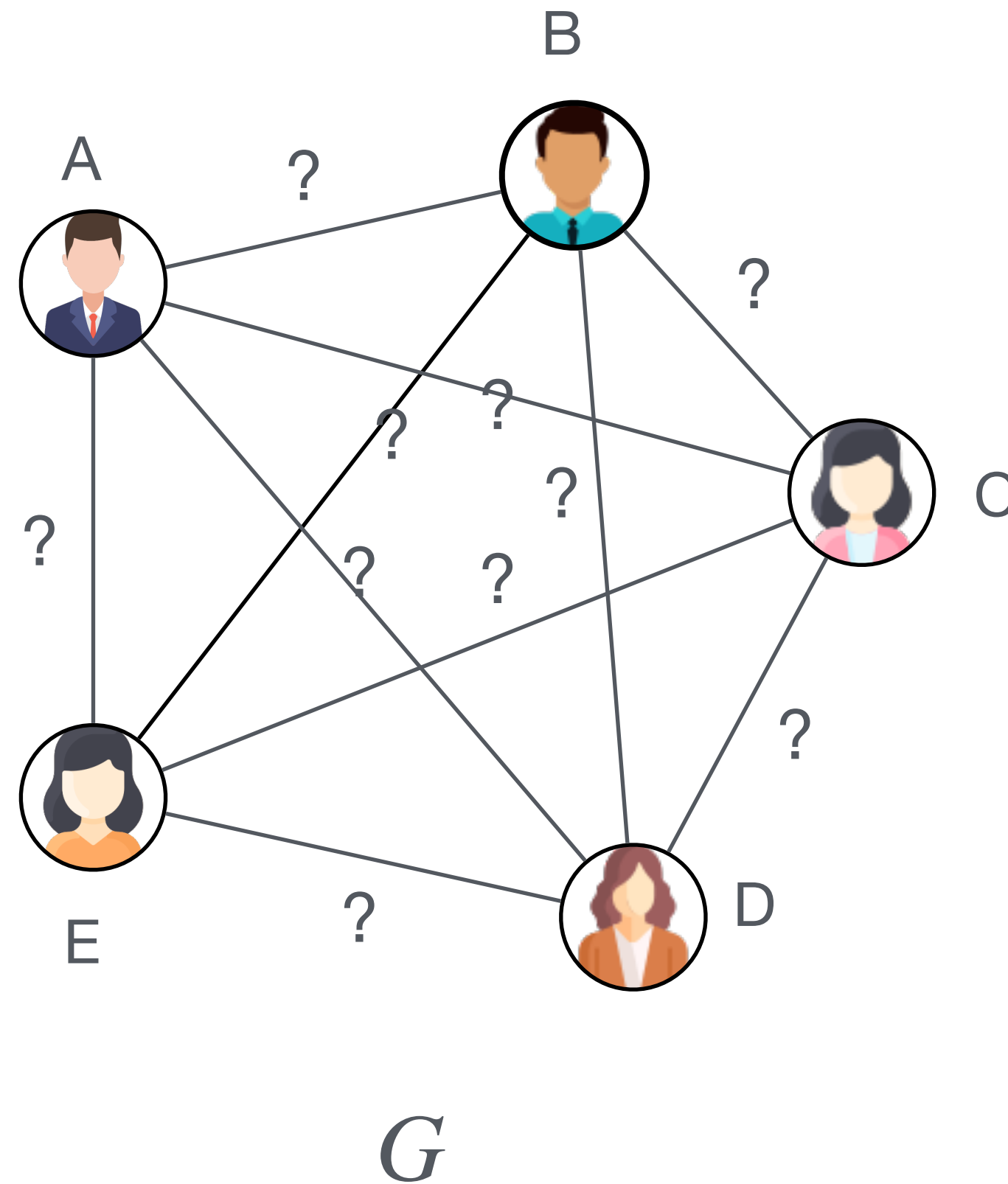
31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining
August 3 - 7, 2025 - Toronto

The limitations of cryptography

Maybe outputs are not that safe

2	2	0	0	0
2	3	0	0	0
0	0	1	1	1
0	0	1	2	2
0	0	1	2	2

G^2



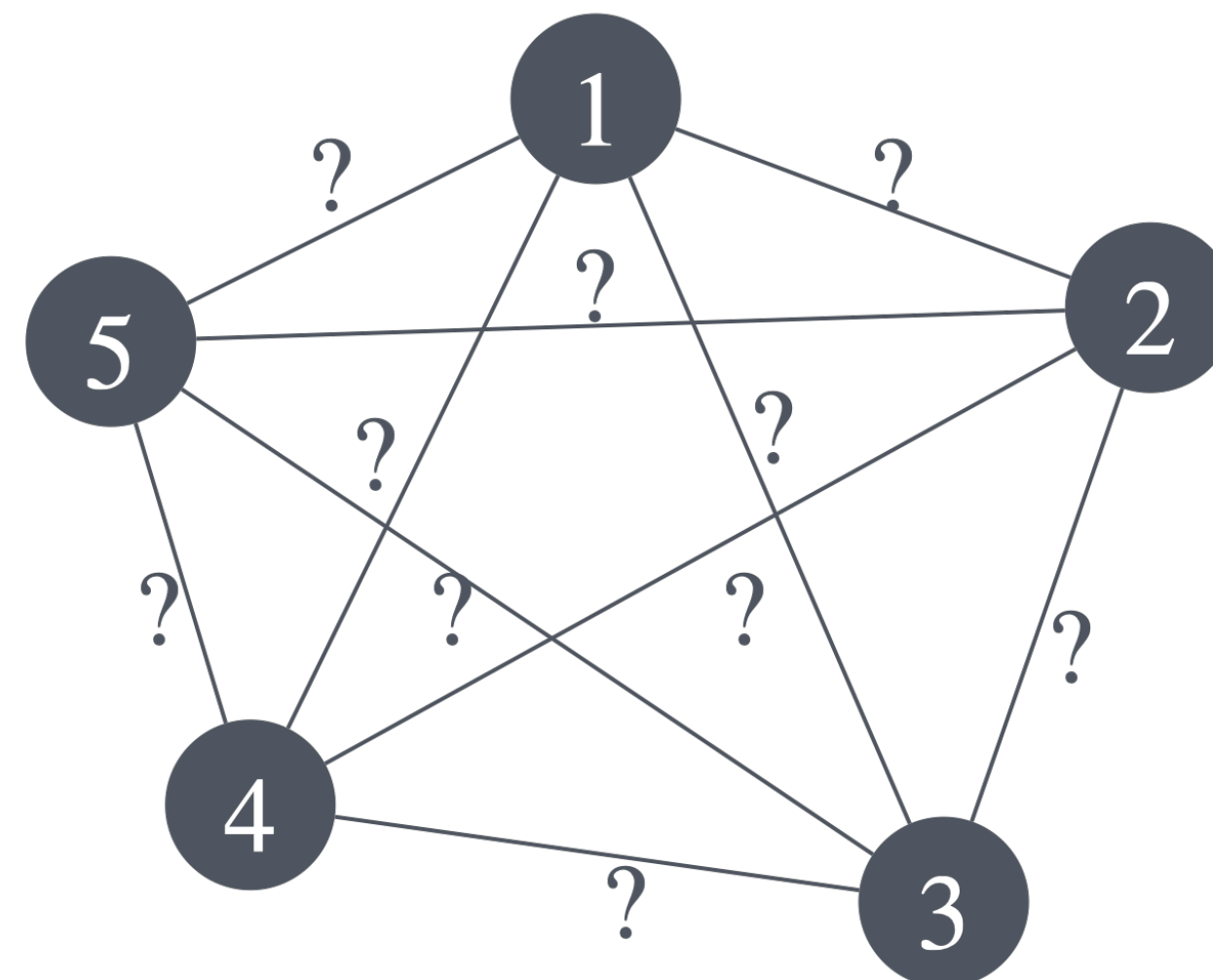
- RQ1 : Can an attacker guess the underlying graph G that produced G^2 ?
- RQ2 : To what extent knowing some existing or non existing edges in G helps the attacker ?

Reconstruction in the topological domain

Initialization

3	1	2	1	1
1	2	1	2	1
2	1	4	1	0
1	2	1	2	1
1	1	0	1	1

G^2



$G^\star$

Topological attacks:

- Degree Combination
- Degree Matching
- Neighbour Matching
- Degree Completion
- Neighbour Completion
- Triangles
- BiCliques

Reconstruction in the spectral domain

Intuition

If $\mathbf{G}$ is the adjacency matrix of an undirected graph :

$$\mathbf{G} = \mathbf{U}\mathbf{\Sigma}\mathbf{V} \text{ (Singular Value Decomposition)}$$

$$\mathbf{G}^2 = \mathbf{U}\mathbf{\Lambda}\mathbf{V} = (\mathbf{U}\mathbf{\Sigma}\mathbf{V})(\mathbf{U}\mathbf{\Sigma}\mathbf{V}) = \mathbf{U}\mathbf{\Sigma}^2\mathbf{V}$$

- $\forall \sigma_i, \exists \lambda_j$ such that $\sigma_i^2 = \lambda_j$
 - Is σ_i positive or negative ?

9	0	0	0
0	4	0	0
0	0	1	0
0	0	0	1

$\mathbf{\Lambda}$

3	0	0	0
0	2	0	0
0	0	1	0
0	0	0	1

$$\mathbf{\Sigma} = \text{diag}(3, 2, 1, 1)$$

...

2	0	0	0
0	1	0	0
0	0	1	0
0	0	0	-3

$$\mathbf{\Sigma} = \text{diag}(2, 1, 1, -3)$$

Reconstruction in the spectral domain

Solution

A greedy optimization algorithm :

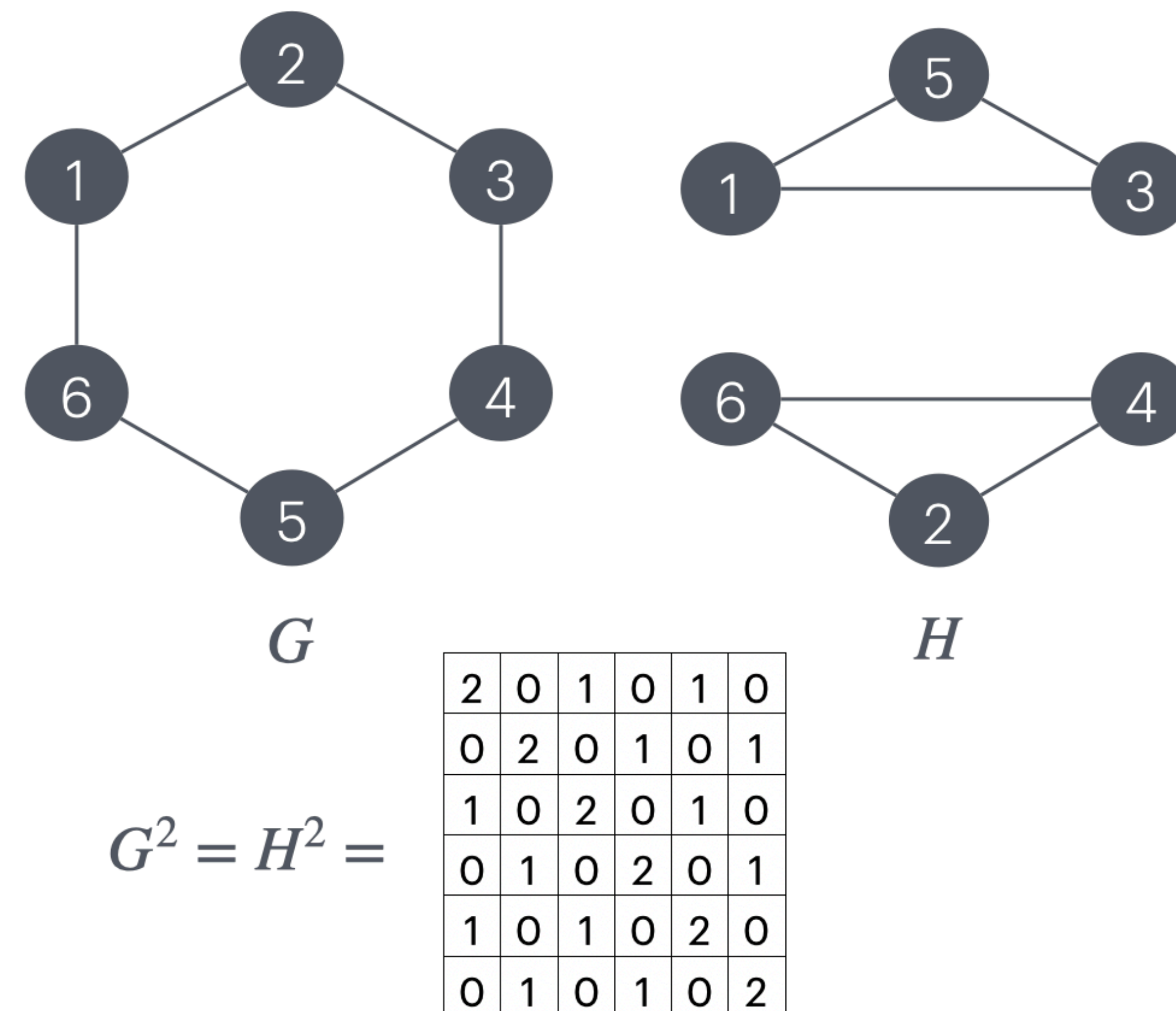
- For each σ_i , reconstruct the graph using $\sigma_i = \sqrt{\lambda_i}$ and $\sigma_i = -\sqrt{\lambda_i}$
- σ_i gets assigned the value that produces a graph that is closest to a binary graph and to the known adjacency information

$$\min_{\sigma \in \{\sqrt{\lambda_i}, -\sqrt{\lambda_i}\}} \alpha \left\| G_{\sigma}^{\star} - BG_{\sigma}^{\star} \right\|_F + \beta \left\| G_{\sigma}^{\star} - G_t^{\star} \right\|_F$$

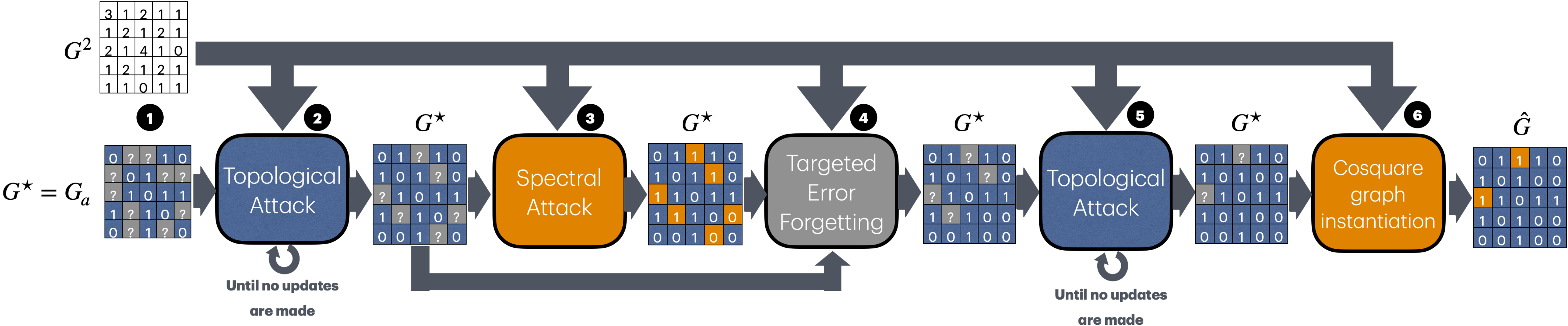
GRAND

Cosquare graphs

- Some graphs have the same common neighbour matrix
- We introduce the notion of cosquare graphs to denote non isomorphic graphs that do not have the same common neighbour matrix



GRAND Pipeline



GRAND

Results

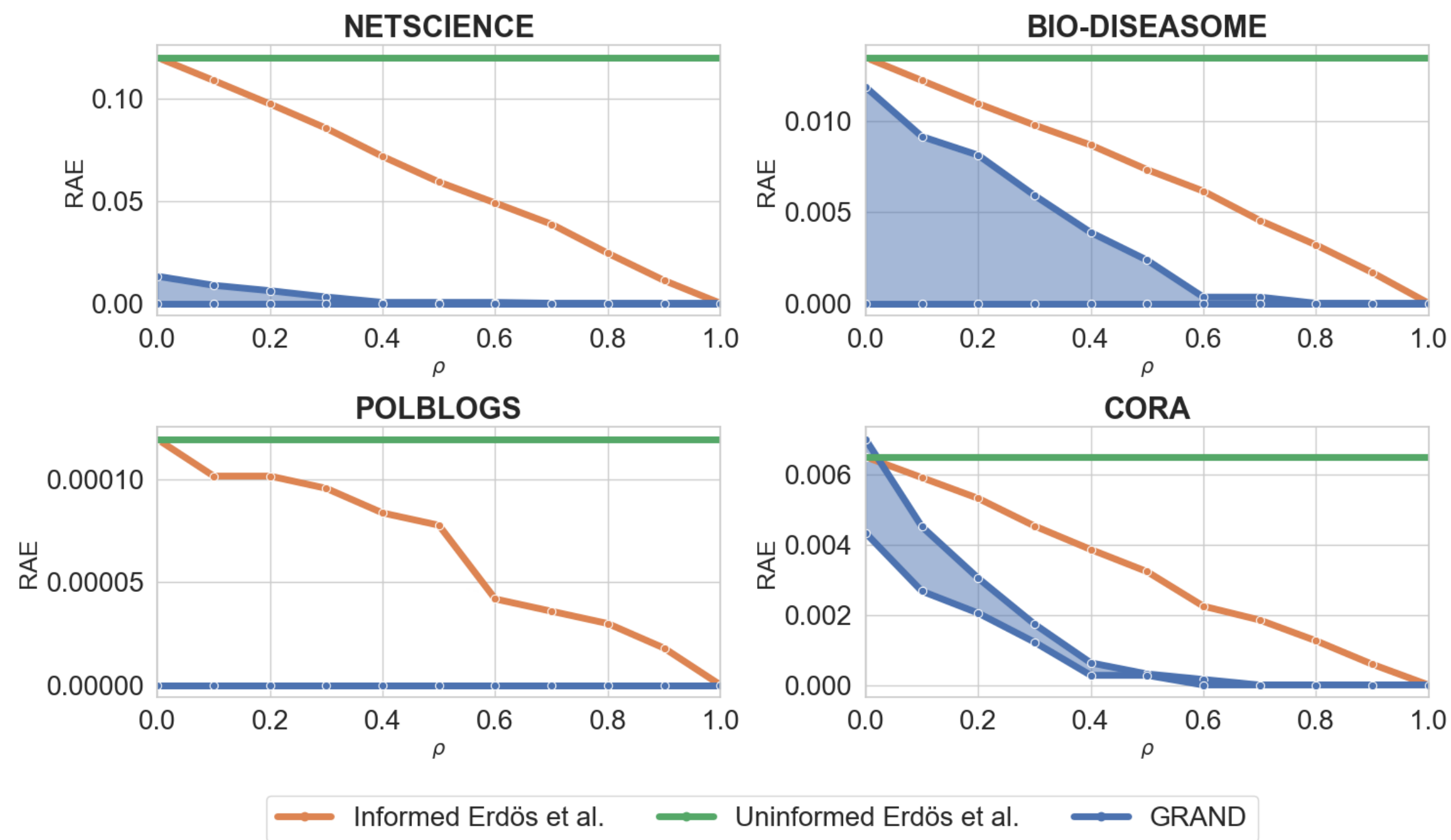


Fig 1. Relative Absolute Error for various datasets and prior knowledge

- Prior knowledge is helpful to reconstruct the graph. The more prior knowledge we have, the more we can reconstruct.
- Perfect reconstruction is sometimes possible, even with no prior knowledge

Cross cutting-insights

- Secure computation is great at what it does : computing on data without directly revealing it
- When the result of the computation is to be disclosed, even if it seems to not reveal the input, special care needs to be taken
- When secure computation needs to be done on the mix of private data from different stakeholders, what each knows about the other matters a lot

Relevance to power systems

- Machine learning models are increasingly applied to the grid (demand forecasting, expansion forecasting, intrusion detection, etc)
- These models create a new attack surface
 - collaboration between semi-trusted grid operators
 - outsourcing of training in the cloud
 - sensitive data reconstruction from model outputs

Relevance to power systems

The GridFM project

- Gathers actors of academia and industry to train Foundation Models on grid data for various use cases (demand forecasting, failure prediction, security, etc)
- They are thinking about adopting the Advanced Privacy-Preserving Federated Learning framework (APPFL) which does secure federated learning (Differential privacy, secure aggregation)
 - To what extent the current solutions are vulnerable to reconstruction attacks from the aggregated results ?
 - Can we build efficient privacy-preservation mechanisms for collaborative grid learning ?

Thank you !

Happy to take your questions



Link to this presentation:

<https://birbaubin.github.io/assets/pdf/hitachi.pdf>



Link to Crypto'Graph paper

<https://dl.acm.org/doi/abs/10.1145/3626232.3653257>



Link to GRAND paper

<https://dl.acm.org/doi/10.1145/3711896.3736988>